The SMUAS method of Knobloch (Numer. Algorithms 2023): implementation check, algorithm and reproduction of the numerical results

galerkit

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Abstract

P. Knobloch, *An algebraically stabilized method for convection–diffusion–reaction problems with optimal experimental convergence rates on general meshes*, Numer. Algorithms 94 (2023) [1], analyses why algebraic stabilizations lose accuracy on meshes without local symmetry and proposes the Symmetrized Monotone Upwind-type Algebraically Stabilized (SMUAS) method. We check that galerkit implements all four methods of the paper (the AFC schemes with the Kuzmin and the BJK limiter, MUAS and SMUAS) together with the limiter variants used in its analysis, state the SMUAS method as an algorithm, and recompute the paper's numerical results: Tables 1–9, the statements of Sections 5 and 6, and Figures 3, 4, 5, 7, 8, 9, 11 and 13. Of the 159 values of Tables 1–9 ($ne = 16, \ldots, 512$, Table 5 up to $ne = 256$; up to $5.3 \cdot 10^5$ unknowns), all agree with the paper in all printed digits.

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1 Is the method implemented?

Answer: yes. All four methods of the paper were already implemented in galerkit, and Tables 1, 2, 4, 5 (first row) and 8 had been checked before. The SMUAS implementation (limiter `"smuas"` in `src/galerkit/convection_diffusion/afc.py`) was checked line by line against (61)–(68) of [1]: the sums in $P_i^{\pm}$ run over $j \in S_i$ with $a_{ij} > 0$ or $a_{ji} > 0$ and those in $Q_i^{\pm}$ over all $j \in S_i$; the reflected values are $u_{ij} = u_i + \nabla u_h|_{T_{ij}} \cdot (x_i - x_j)$ with T_{ij} the simplex of the patch entered by the half line $x_i + t(x_i - x_j)$, $t > 0$; $R_i^{\pm} = 1$ at Dirichlet nodes and where $P_i^{\pm} = 0$; β_{ij} and b_{ij} follow (36), (31),

Table I: What [1] uses and where it is in galerkit (`src/galerkit/convection_diffusion`). "new" marks what was added for this reproduction.

item of the paper	galerkit
AFC, Kuzmin limiter (19)–(23), $R_i^{\pm}=1$ at Dirichlet nodes	`stab="afc_kuzmin"` (limiter `"kuzmin"`)
AFC, BJK limiter (25)–(28) with $a_{ji}:=0$ at Dirichlet nodes	`stab="afc"`, μ_i = `afc_gamma` (a number, or `"bjk"` for (30))
MUAS (31)–(36)	`stab="muas"`
SMUAS (60)–(64) with (67) or (68)	`stab="smuas"`, `smuas_weights="67"|"68"`
Kuzmin with $P_i^{\pm}$ from (25) (Fig. 7, right)	new: `AFCOperator(kuzmin_p="25")`
$R_i^{\pm}=\min\{1,\mu_i Q_i^{\pm}/P_i^{\pm}\}$, (27), (47), (48)	new: `AFCOperator(r_mu=`scalar or per node`)`
Example 1	`bjk_polynomial`
Examples 2–5	new: `kn23_ex2`, …, `kn23_ex5` (Ex. 3 in an overflow-free form)
Grids 1, 4, 5; 6, 7, 8	`rectangle_mesh` (`right`, `rows`, `left`, `alternating`, `crossed`), `distorted_rows_mesh`
Grids 2, 3	from the BJK16 reproduction (Grid 3 read from the vector figure of [2])
Grid 9	new: `kn23_grids.grid9`
Table 9 grid (shift $1.5h$, last node $0.75h$)	new: `distorted_rows_mesh(last_shift=0.75)`
Table 10: triangulated FVCA5 `mesh4_1` family	*not available* (see Section 7)
nonlinear solver "fixed point rhs" (Remark 5)	`afc_solver="fp_rhs"`; also `"newton"`, `"fp_anderson"`

(32). For this reproduction we added the limiter variants studied in Section 5 of the paper ($P_i^{\pm}$ from (25) for the Kuzmin limiter, and the factor μ_i of (27)/(47) in $R_i^{\pm}$), Examples 2–5, Grids 6–9 and the grid of Table 9.

2 The SMUAS algorithm

Setting. P_1 elements on a simplicial triangulation of $\Omega\subset\mathbb{R}^d$, nodes $x_1,\dots,x_N$ with $x_{M+1},\dots,x_N$ on $\partial\Omega$, S_i the neighbours of x_i, Δ_i the patch of x_i, and $A=(a_{ij})$ the Galerkin matrix of $-\varepsilon\Delta u+\boldsymbol{b}\cdot\nabla u+cu=g$, $a_{ij}=a(\varphi_j,\varphi_i)$. The method solves

$$\sum_{j=1}^{N}\big(a_{ij}+b_{ij}(U)\big)u_j=g_i,\quad i\le M,\qquad u_i=u_b(x_i),\quad i>M, \tag{1}$$

with the symmetric, row-sum-free stabilization matrix $b_{ij}(U)=-\max\{\beta_{ij}(U)a_{ij},0,\beta_{ji}(U)a_{ji}\}$ ($i\neq j$), $b_{ii}=-\sum_{j\neq i}b_{ij}$.

Algorithm SMUAS. *Input:* A, g, u_b, the mesh, the choice (67) or (68) of the weights, a tolerance τ. *Output:* U solving (1).

Set-up (once per mesh).

S1. For every interior node i and every $j\in S_i$: find the simplex $T_{ij}\subset\Delta_i$ entered by the half line $\{x_i+t(x_i-x_j):t>0\}$ (the simplex whose barycentric gradients satisfy $\nabla\lambda_k\cdot(x_i-x_j)\ge 0$

for all its vertices $x_k \neq x_i$). Store the row $w_{ij} = e_i + \sum_{x_k \in T_{ij}} (\nabla \lambda_k \cdot (x_i - x_j))\, e_k$, so that the reflected value is $u_{ij} = w_{ij} \cdot U = u_i + \nabla u_h|_{T_{ij}} \cdot (x_i - x_j)$ (60)

S2. Weights, (67): $p_{ij} = \max\{a_{ij}, 0, a_{ji}\}$, $q_{ij} = \max\{|a_{ij}|, a_{ji}\}$; or (68): $p_{ij} = q_{ij} = 1$. Set $p_{ij} = 0$ unless $a_{ij} > 0$ or $a_{ji} > 0$.

Stabilization matrix $B(U)$ for a given U.

B1. For $i \leq M$ and $j \in S_i$: $\delta_{ij} = u_i - u_j$ and $r_{ij} = u_i - u_{ij}$.

B2. $P_i^{\pm} = \sum_{j \in S_i} p_{ij}\{(\delta_{ij})^{\pm} + (r_{ij})^{\pm}\}$, $Q_i^{\pm} = \sum_{j \in S_i} q_{ij}\{(-\delta_{ij})^{\pm} + (-r_{ij})^{\pm}\}$ (61)–(64) with $z^+ = \max\{0, z\}$, $z^- = \min\{0, z\}$.

B3. $R_i^{\pm} = \min\{1, Q_i^{\pm}/P_i^{\pm}\}$, and $R_i^{\pm} = 1$ if $P_i^{\pm} = 0$ or $i > M$. (20), (21)

B4. $\beta_{ij} = 1 - R_i^+$ if $u_i > u_j$, $\beta_{ij} = 1 - R_i^-$ if $u_i < u_j$, $\beta_{ij} = 0$ if $u_i = u_j$. (36)

B5. $b_{ij} = -\max\{\beta_{ij} a_{ij}, 0, \beta_{ji} a_{ji}\}$ for every edge, $b_{ii} = -\sum_{j \neq i} b_{ij}$. (31), (32)

Nonlinear solve. Let D be B with all $\beta_{ij} = 1$, i.e. $d_{ij} = -\max\{a_{ij}, 0, a_{ji}\}$.

N1. U^0: solve $(A + D)U^0 = g$ (with the Dirichlet values), the monotone low-order solution.

N2. For $k = 0, 1, \ldots$: compute $B(U^k)$ by B1–B5 and the residual $r^k = g - (A + B(U^k))U^k$ on the free nodes; stop if $\|r^k\|_2 \leq \tau$.

N3. *Fixed point rhs* (used in [1], Remark 5): solve $(A + D)\tilde{U} = g - (B(U^k) - D)U^k$ with the factorization of $A+D$ from N1, and set $U^{k+1} = U^k + \omega_k(\tilde{U} - U^k)$ with a damping $\omega_k \in (0, 1]$; or *semismooth Newton*: $U^{k+1} = U^k - \lambda_k J(U^k)^{-1} r^k$ with the generalized Jacobian of the piecewise smooth map $U \mapsto (A + B(U))U$ and an Armijo step λ_k.

Properties (proved in [1] for arbitrary simplicial meshes). (A1) and hence existence of a solution (Theorem 4), the local DMP via (A2) (Theorem 5), and linearity preservation (Remark 2: for $u_h \in P_1(\mathbb{R}^d)$ one has $r_{ij} = -\delta_{ij}$, hence $P_i^+ = -P_i^- = \sum_j p_{ij}|u_i - u_j| \leq \sum_j q_{ij}|u_i - u_j| = Q_i^+ = -Q_i^-$ by $p_{ij} \leq q_{ij}$, so $R_i^{\pm} = 1$ and $B(U) = 0$). The cost per evaluation of $B(U)$ is one sparse product WU (the reflected values) plus edge loops, i.e. linear in the number of edges; the set-up S1 is a single pass over the patches.

Implementation in galerkit.

- S1: `reflection_operator`, a sparse matrix W with one row per directed edge;
- B1–B5: `AFCOperator.b_edges`;
- N1–N3: `solve_afc` with `afc_solver` set to `"fp_rhs"`, `"fp_anderson"` (Anderson acceleration) or `"newton"` (exact generalized Jacobian of SMUAS, `_jacobian_muas`).

3 Set-up of the computations

- P_1 elements, no lumping of the reaction term (in contrast to [2]), ε as stated.

- ne is the number of edges on the part of $\partial\Omega$ on $y = 1$: Grids 1, 4–8 have $ne \times ne$ squares; Grid 2 has $ne = 2$ and Grid 3 $ne = 4$ on level 0 and is refined uniformly, so that grids with the same ne have similar numbers of triangles; Grid 9 is built for each ne.
- Error norm $\|v\|_h^2 = \varepsilon|v|_1^2 + \sigma_0\|v\|_0^2 + b_h(u_h; v, v)$ with $v = u - u_h$, the last term evaluated at $i_h u - u_h$, and $\sigma_0 = 1$. For Example 2, where $c = 0$, the paper's Table 3 is reproduced only with $\sigma_0 = 1$, i.e. the L^2 term is kept although $\sigma_0 = 0$ there: $\sqrt{\varepsilon|e|_1^2 + b_h} = 8.560 \cdot 10^{-3}$ against the printed $1.179 \cdot 10^{-2} = \sqrt{(8.560 \cdot 10^{-3})^2 + \|e\|_0^2}$ for $ne = 16$.
- Nonlinear solver: semismooth Newton with the exact generalized Jacobian (Kuzmin, MUAS, SMUAS) or with coloured finite differences (the variants), stopped at a Euclidean residual of 10^{-10}; where Newton is slow, the Anderson-accelerated fixed point rhs. The paper used fixed point rhs; the values do not depend on the solver once the residual is small.
- "Odd horizontal grid lines" in (47), (48) are counted from $y = 0$ as line 1, i.e. $y = 0, 2h, 4h, \ldots$: with μ on these lines Example 2 is solved exactly, with μ on the other lines it is not (error $2.5 \cdot 10^{-2}$).

4 Tables

Table S: Agreement with Tables 1–9 of [1]: entries computed, entries differing in the printed digits, largest relative deviation from the printed values (the printing rounds by up to $5 \cdot 10^{-4}$), and the largest final Euclidean residual of the nonlinear solve.

Table	case	ne	entries	differ	max dev.	residual
1	Example 1, $\varepsilon = 10^{-8}$, Grid 1, AFC with the Kuzmin limiter	16–512	18	0	3.2e-04	6e-14
2	Example 1, $\varepsilon = 10^{-8}$, Grid 4, AFC with the Kuzmin limiter	16–512	18	0	3.1e-04	1e-10
3	Example 2, Grid 4, AFC with the Kuzmin limiter	16–512	18	0	3.7e-04	1e-10
4	Example 1, $\varepsilon = 10^{-8}$, Grid 4, SMUAS with (67)	16–512	18	0	3.0e-04	9e-11
5	Example 1, $\varepsilon = 10^{-8}$, Grid 4, SMUAS with (68)	16–256	15	0	4.6e-04	1e-12
6	Example 1, $\varepsilon = 10^{-8}$, Grid 2, SMUAS with (67)	16–512	18	0	3.7e-04	2e-11
7	Example 1, $\varepsilon = 10^{-8}$, Grid 3, SMUAS with (68)	16–512	18	0	3.4e-04	3e-11
8	Example 1, $\varepsilon = 10$, Grid 5 with shift $0.8h$, SMUAS with (67)	16–512	18	0	4.3e-04	1e-12
9	Example 1, $\varepsilon = 10$, Grid 5 with shift $1.5h$ ($0.75h$ for the last node), SMUAS with (67)	16–512	18	0	2.8e-04	7e-11

Table 1: Example 1, $\varepsilon = 10^{-8}$, Grid 1, AFC with the Kuzmin limiter. galerkit values; "ord." is $\log_2$ of the ratio to the previous row. Every entry agrees with the paper in all printed digits.

ne	$\|u-u_h\|_0$	ord.	$\|u-u_h\|_1$	ord.	$\|u-u_h\|_h$	ord.
16	1.934e−2		4.937e−1		5.007e−2	
32	5.359e−3	1.85	2.305e−1	1.10	1.149e−2	2.12
64	1.385e−3	1.95	1.082e−1	1.09	2.649e−3	2.12
128	3.442e−4	2.01	5.154e−2	1.07	6.152e−4	2.11
256	8.536e−5	2.01	2.566e−2	1.01	1.586e−4	1.96
512	2.126e−5	2.01	1.342e−2	0.93	3.876e−5	2.03

Table 2: Example 1, $\varepsilon = 10^{-8}$, Grid 4, AFC with the Kuzmin limiter. galerkit values; "ord." is $\log_2$ of the ratio to the previous row. Every entry agrees with the paper in all printed digits.

ne	$\|u-u_h\|_0$	ord.	$\|u-u_h\|_1$	ord.	$\|u-u_h\|_h$	ord.
16	2.019e−2		6.005e−1		5.663e−2	
32	6.285e−3	1.68	4.832e−1	0.31	2.138e−2	1.41
64	2.308e−3	1.45	4.549e−1	0.09	9.485e−3	1.17
128	1.092e−3	1.08	4.442e−1	0.03	4.490e−3	1.08
256	5.543e−4	0.98	4.368e−1	0.02	2.187e−3	1.04
512	2.823e−4	0.97	4.327e−1	0.01	1.083e−3	1.01

Table 3: Example 2, Grid 4, AFC with the Kuzmin limiter. galerkit values; "ord." is $\log_2$ of the ratio to the previous row. Every entry agrees with the paper in all printed digits.

ne	$\|u-u_h\|_0$	ord.	$\|u-u_h\|_1$	ord.	$\|u-u_h\|_h$	ord.
16	8.104e−3		4.401e−1		1.179e−2	
32	4.291e−3	0.92	4.700e−1	-0.09	6.227e−3	0.92
64	2.204e−3	0.96	4.851e−1	-0.05	3.157e−3	0.98
128	1.117e−3	0.98	4.926e−1	-0.02	1.580e−3	1.00
256	5.618e−4	0.99	4.963e−1	-0.01	7.893e−4	1.00
512	2.817e−4	1.00	4.982e−1	-0.01	3.974e−4	0.99

Table 4: Example 1, $\varepsilon = 10^{-8}$, Grid 4, SMUAS with (67). galerkit values; "ord." is $\log_2$ of the ratio to the previous row. Every entry agrees with the paper in all printed digits.

ne	$\|u-u_h\|_0$	ord.	$\|u-u_h\|_1$	ord.	$\|u-u_h\|_h$	ord.
16	2.147e−2		4.734e−1		5.530e−2	
32	6.353e−3	1.76	2.529e−1	0.90	1.479e−2	1.90
64	1.783e−3	1.83	1.363e−1	0.89	3.922e−3	1.92
128	4.706e−4	1.92	7.220e−2	0.92	1.054e−3	1.90
256	1.221e−4	1.95	3.807e−2	0.92	2.940e−4	1.84
512	3.135e−5	1.96	2.002e−2	0.93	7.896e−5	1.90

Table 5: Example 1, $\varepsilon = 10^{-8}$, Grid 4, SMUAS with (68). galerkit values; "ord." is $\log_2$ of the ratio to the previous row. Every entry agrees with the paper in all printed digits.

ne	$\|u-u_h\|_0$	ord.	$\|u-u_h\|_1$	ord.	$\|u-u_h\|_h$	ord.
16	2.208e−2		4.748e−1		5.702e−2	
32	6.605e−3	1.74	2.515e−1	0.92	1.530e−2	1.90
64	1.860e−3	1.83	1.336e−1	0.91	4.008e−3	1.93
128	4.924e−4	1.92	6.959e−2	0.94	1.046e−3	1.94
256	1.279e−4	1.95	3.635e−2	0.94	2.823e−4	1.89

Table 6: Example 1, $\varepsilon = 10^{-8}$, Grid 2, SMUAS with (67). galerkit values; "ord." is $\log_2$ of the ratio to the previous row. Every entry agrees with the paper in all printed digits.

ne	$\|u-u_h\|_0$	ord.	$\|u-u_h\|_1$	ord.	$\|u-u_h\|_h$	ord.
16	9.950e−3		2.432e−1		1.956e−2	
32	2.764e−3	1.85	1.125e−1	1.11	4.638e−3	2.08
64	7.357e−4	1.91	5.210e−2	1.11	1.139e−3	2.03
128	1.889e−4	1.96	2.461e−2	1.08	2.812e−4	2.02
256	4.778e−5	1.98	1.189e−2	1.05	6.888e−5	2.03
512	1.202e−5	1.99	5.837e−3	1.03	1.709e−5	2.01

Table 7: Example 1, $\varepsilon = 10^{-8}$, Grid 3, SMUAS with (68). galerkit values; "ord." is $\log_2$ of the ratio to the previous row. Every entry agrees with the paper in all printed digits.

ne	$\|u-u_h\|_0$	ord.	$\|u-u_h\|_1$	ord.	$\|u-u_h\|_h$	ord.
16	2.218e−2		4.338e−1		5.994e−2	
32	7.257e−3	1.61	2.175e−1	1.00	1.637e−2	1.87
64	2.093e−3	1.79	1.009e−1	1.11	4.066e−3	2.01
128	5.703e−4	1.88	4.635e−2	1.12	1.000e−3	2.02
256	1.520e−4	1.91	2.184e−2	1.09	2.524e−4	1.99
512	4.046e−5	1.91	1.046e−2	1.06	6.448e−5	1.97

Table 8: Example 1, $\varepsilon = 10$, Grid 5 with shift $0.8h$, SMUAS with (67). galerkit values; "ord." is $\log_2$ of the ratio to the previous row. Every entry agrees with the paper in all printed digits.

ne	$\|u-u_h\|_0$	ord.	$\|u-u_h\|_1$	ord.	$\|u-u_h\|_h$	ord.
16	3.155e−2		5.855e−1		1.976e0	
32	7.267e−3	2.12	3.002e−1	0.96	9.676e−1	1.03
64	1.665e−3	2.13	1.518e−1	0.98	4.826e−1	1.00
128	4.111e−4	2.02	7.642e−2	0.99	2.420e−1	1.00
256	1.048e−4	1.97	3.837e−2	0.99	1.214e−1	1.00
512	2.659e−5	1.98	1.922e−2	1.00	6.080e−2	1.00

Table 9: Example 1, $\varepsilon = 10$, Grid 5 with shift $1.5h$ ($0.75h$ for the last node), SMUAS with (67). galerkit values; "ord." is $\log_2$ of the ratio to the previous row. Every entry agrees with the paper in all printed digits.

ne	$\|u-u_h\|_0$	ord.	$\|u-u_h\|_1$	ord.	$\|u-u_h\|_h$	ord.
16	7.338e−2		8.767e−1		3.620e0	
32	2.287e−2	1.68	4.854e−1	0.85	1.732e0	1.06
64	5.395e−3	2.08	2.506e−1	0.95	8.234e−1	1.07
128	1.319e−3	2.03	1.281e−1	0.97	4.091e−1	1.01
256	3.541e−4	1.90	6.501e−2	0.98	2.062e−1	0.99
512	9.619e−5	1.88	3.275e−2	0.99	1.037e−1	0.99

5 The statements of Sections 5 and 6

Table C1: Example 2, AFC with the Kuzmin limiter on Grid 4: in the interior, $u_h = x + \alpha$ on the odd and $u_h = x - \beta$ on the even horizontal grid lines, (40), (41); α, β are medians over $[0.2, 0.8]^2$. $\alpha + \beta$ against δ of (46), and the maximal nodal error.

ne	α	β	$\alpha+\beta$	$\delta = h^2/(2h-3\varepsilon)$	$\max\|u_h-u\|$
20	1.2004e-02	1.2947e-02	2.4952e-02	2.5000e-02	1.532e-02
40	6.0052e-03	6.4946e-03	1.2500e-02	1.2500e-02	7.717e-03
80	3.0026e-03	3.2473e-03	6.2500e-03	6.2500e-03	3.879e-03

- Example 2 with the Kuzmin limiter modified by (47), (48) on the odd lines (Grid 4, $ne = 20$): maximal nodal error 3.2e-08 (final residual 1e-10), i.e. the exact solution is recovered, as stated in Section 5.
- SMUAS with (67) on Example 2: maximal nodal errors on Grids 1–9 ($ne = 16$): 7e-16 1e-08 2e-08 2e-12 7e-14 7e-16 1e-15 2e-15 1e-08; the method is exact on every grid (the larger values are the tolerance of the nonlinear solve).
- Example 3, Kuzmin on Grid 1 ($ne = 20$): nodal error 9e-14 except on the last interior vertical line, where it is 0.029 (Fig. 7, left).
- Example 3, SMUAS on Grid 4 ($ne = 20$): with (68) the nodal error is below 10^{-9} except on the last interior vertical line, as stated in Section 6. With (67) the solution has an error of 1.1e-03 also on the second-last line.

Table C2: Example 5: the convergence statements of Sections 5 and 6 (no table in the paper).

method	ne	$\|e\|_0$	$\|e\|_1$	$\|e\|_h$	orders		
Kuzmin, Grid 1	16	3.299e−4	3.724e−2	3.299e−4			
	32	8.247e−5	1.862e−2	8.249e−5	2.00	1.00	2.00
	64	2.062e−5	9.307e−3	2.064e−5	2.00	1.00	2.00
	128	5.156e−6	4.653e−3	5.177e−6	2.00	1.00	2.00
Kuzmin, Grid 4	16	2.310e−2	8.354e−1	3.454e−2			
	32	1.169e−2	8.277e−1	1.826e−2	0.98	0.01	0.92
	64	5.893e−3	8.255e−1	9.434e−3	0.99	0.00	0.95
	128	2.960e−3	8.246e−1	4.798e−3	0.99	0.00	0.98
SMUAS (67), Grid 4	16	1.617e−3	5.646e−2	1.827e−3			
	32	4.130e−4	2.684e−2	4.705e−4	1.97	1.07	1.96
	64	1.042e−4	1.292e−2	1.186e−4	1.99	1.05	1.99
	128	2.616e−5	6.313e−3	2.972e−5	1.99	1.03	2.00

6 Figures

All figures are wire frames of u_h along the grid lines parallel to the axes, as in the paper; they are qualitative and agree with the paper in every feature described there.

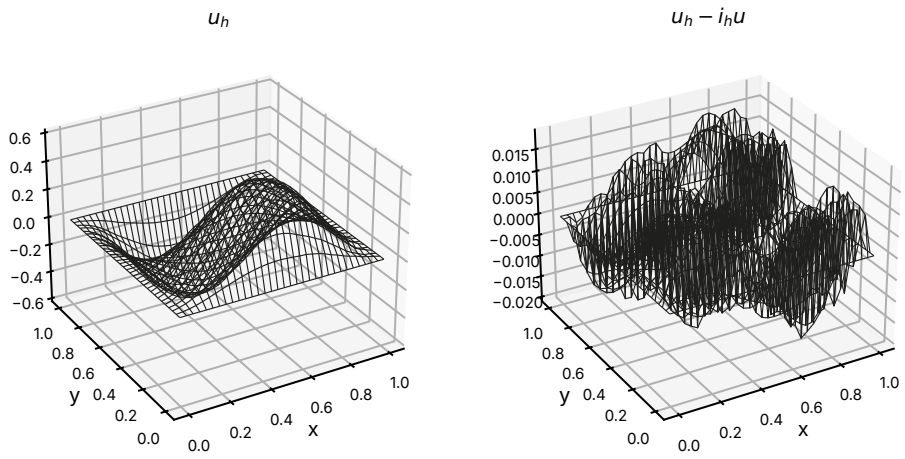


Figure 1: (Fig. 3 of [1]) Example 1, $\varepsilon = 10^{-8}$, Kuzmin, Grid 4, $ne = 32$: u_h and $u_h - i_h u$.

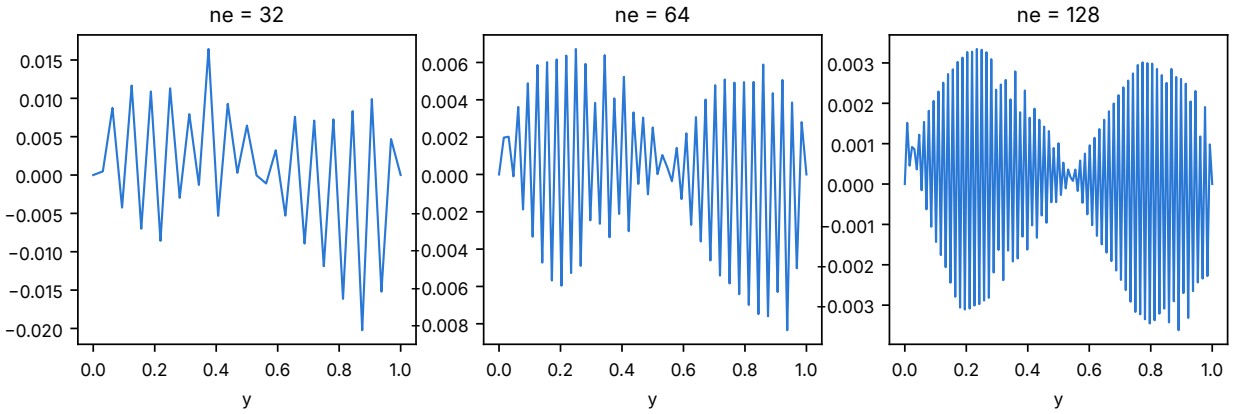


Figure 2: (Fig. 4) $u_h - i_h u$ along $x = 0.25$, $ne = 32, 64, 128$. The amplitudes (≈ 0.02, 0.007, 0.003) match the vertical scales of the paper's figure.

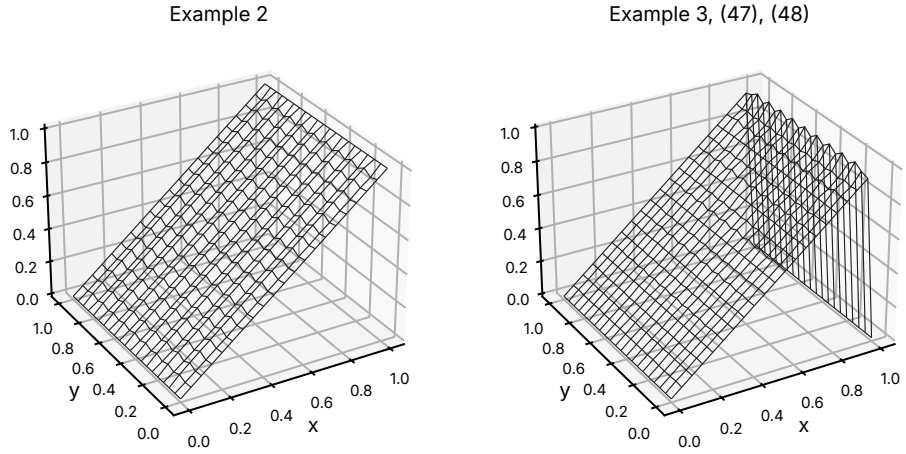


Figure 3: (Fig. 5) Kuzmin, Grid 4, $ne = 20$: Example 2, and Example 3 with (47), (48).

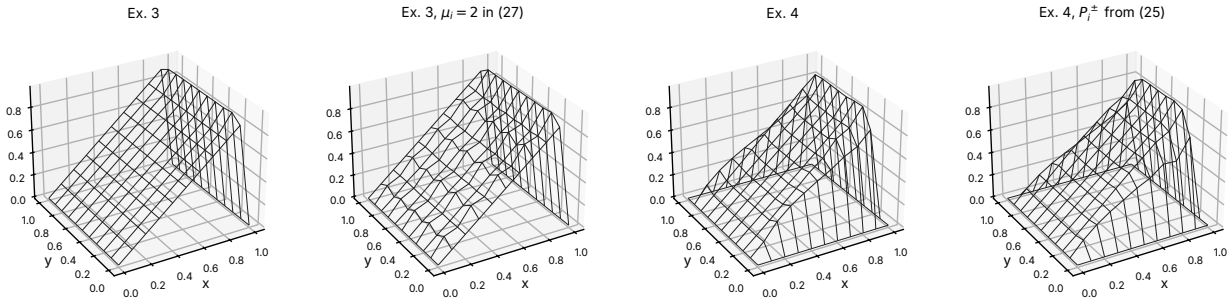


Figure 4: (Fig. 7) Kuzmin, Grid 1, $ne = 10$: Example 3; Example 3 with $\mu_i = 2$ in (27); Example 4; Example 4 with $P_i^{\pm}$ from (25).

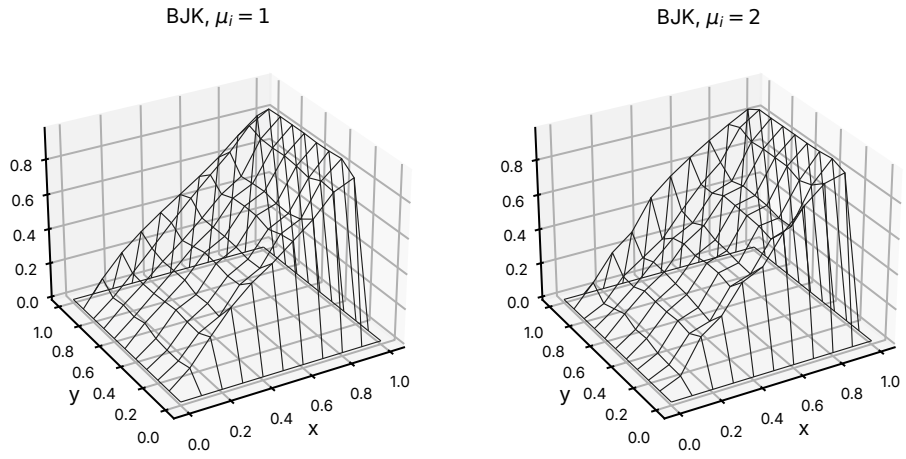


Figure 5: (Fig. 8) Example 4, BJK limiter, Grid 1, $ne = 10$, $\mu_i = 1$ and $\mu_i = 2$.

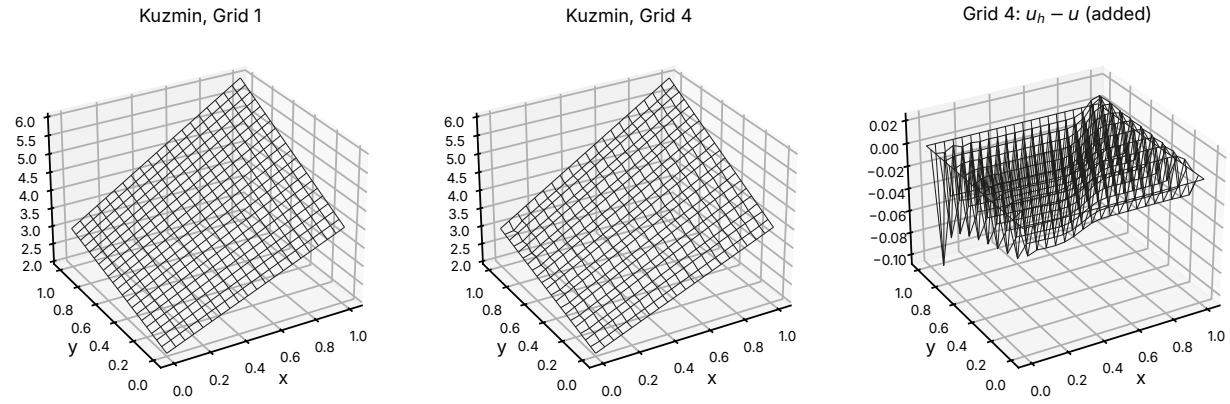


Figure 6: (Fig. 9) Example 5, Kuzmin, $ne = 20$, Grids 1 and 4; the third panel (not in the paper) shows $u_h - u$ on Grid 4, where the oscillation is clearer than in the solution itself.

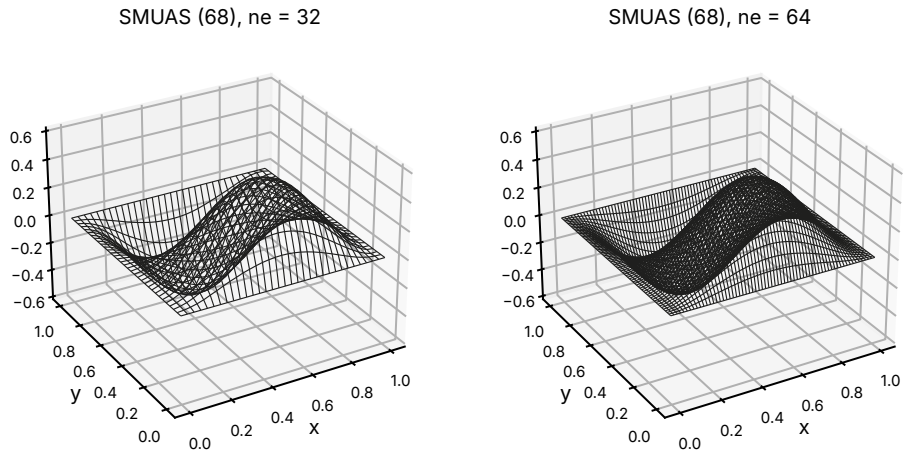


Figure 7: (Fig. 11) Example 1, $\varepsilon = 10^{-8}$, SMUAS (68), Grid 4, $ne = 32$ and 64; compare Fig. 3.

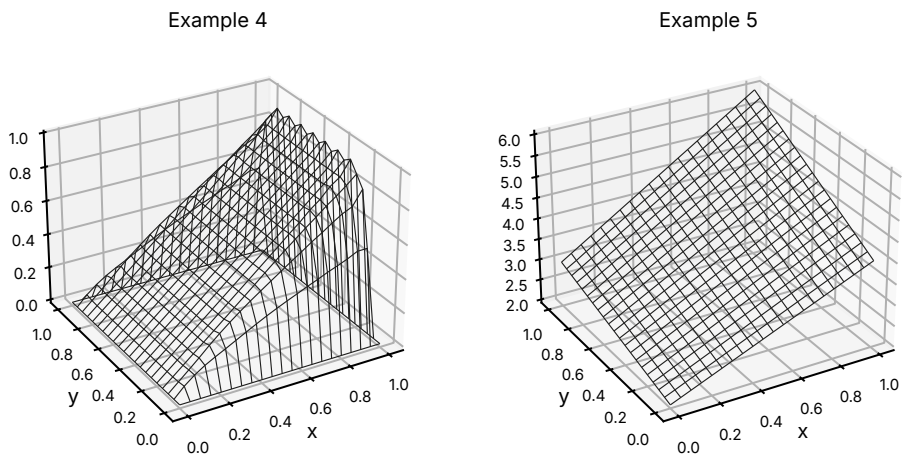


Figure 8: (Fig. 13) SMUAS (67), Grid 4, $ne = 20$: Examples 4 and 5.

7 Discussion

Agreement. Table S shows the agreement for every recomputed table; together with the checks of Section 5 and the figures, every numerical statement of [1] that does not depend on the FVCA5 meshes is reproduced (Table 5 up to $ne = 256$). Three points of the set-up had to be inferred and are now fixed by the data: the $\sigma_0 = 1$ in $\|\cdot\|_h$ also for $c = 0$ (Table 3), the counting of the "odd horizontal grid lines" in (47), (48), and the use of the (exact, recovered) Grids 2 and 3 of [2].

One nuance. For Example 3 on Grid 4 the paper states that SMUAS is nodally exact except on the last interior vertical grid line. This holds for the weights (68). With (67) the computed solution (residual $4 \cdot 10^{-11}$) also has an error of about 10^{-3} on the second-last line; the paper does not say which weights it used for this statement, and its remark that (67) and (68) give "similar" results remains true in the norms.

Table 10 not recomputed. Table 10 uses a triangulated version of the `mesh4_1` family of the FVCA5 benchmark. The meshes are distributed only by the benchmark web site, which this environment cannot reach, and only the coarsest of the six meshes is drawn in the paper (Fig. 12, right); the finer meshes cannot be inferred from it. The run script reads any mesh as a `SimplexMesh`, so Table 10 can be added as soon as the files are available.

Solvers. Semismooth Newton (exact generalized Jacobian for Kuzmin, MUAS and SMUAS) converged in 7–53 iterations for Tables 1, 4 and 6–9 up to $ne = 512$ (up to 40 minutes for the $5.3 \cdot 10^5$ unknowns of Grid 2). For the Kuzmin limiter on Grid 4 (Tables 2 and 3) the Anderson-accelerated fixed point rhs was used at $ne = 256$ and 512 (8 775 and 550 iterations at $ne = 512$), as Newton becomes slow there. Table 5 (SMUAS with (68) on Grid 4) is reported up to $ne = 256$: at $ne = 512$ neither Newton nor the Anderson iteration had converged after 1.5 hours, and the run was stopped. The paper's Remark 5 (fixed point rhs behaves for SMUAS as for the Kuzmin limiter) is therefore confirmed for (67) but not tested at the finest level for (68).

Minimality and the sharper track. The SMUAS properties used here are exactly those of Table 11 of the paper: DMP on arbitrary simplicial meshes (via (A2), which needs only $p_{ij} > 0$

for $a_{ij} > 0$, (66)), linearity preservation (which needs only $p_{ij} \le q_{ij}$, (65)), upwind character (the $\max\{\beta_{ij}a_{ij}, 0, \beta_{ji}a_{ji}\}$ form (31)), and the observed optimal rates. The last property is experimental: no a priori estimate better than the $O(h^{1/2})$ of (16) is proved for SMUAS, and the reproduced Tables 4–9 are the evidence. A proof of optimal-order convergence on general meshes (the open problem stated at the end of [1]) would be the sharper result; the linearity-preserving BJK limiter [3] has the DMP and linearity preservation as well but not the upwind character, and its nonlinear problems are much harder to solve.

References

[1] P. Knobloch, An algebraically stabilized method for convection–diffusion–reaction problems with optimal experimental convergence rates on general meshes, *Numer. Algorithms* 94 (2023) 1229–1267. https://doi.org/10.1007/s11075-023-01511-2

[2] G. R. Barrenechea, V. John, P. Knobloch, Analysis of algebraic flux correction schemes, *SIAM J. Numer. Anal.* 54 (2016) 2427–2451.

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[4] V. John, P. Knobloch, On algebraically stabilized schemes for convection–diffusion–reaction problems, *Numer. Math.* 152 (2022) 553–585.